

R Practice Project 5 – Counting and Probability

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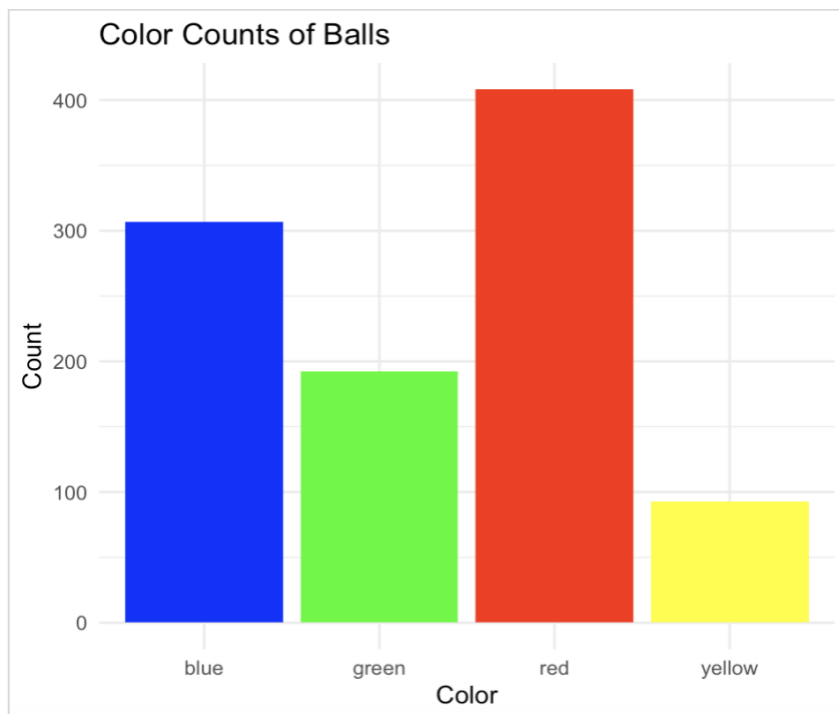
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ALY 6000 Introduction to Analytics

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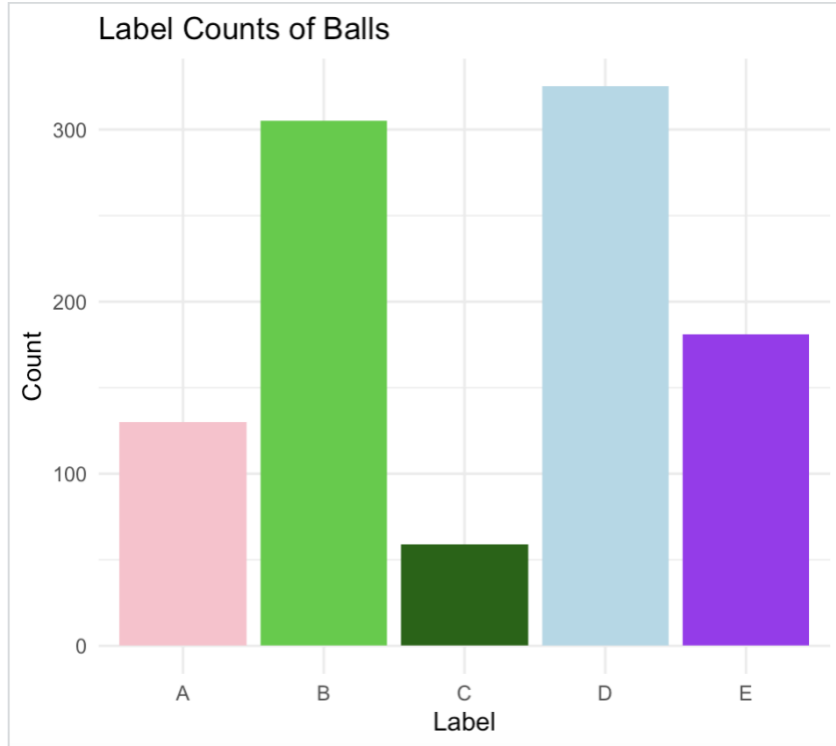
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The purpose of ALY6000 Module 5 project was to examine a dataset on ball colors (red, blue, green, yellow), and labels (A, B, C, D, E) for each of these balls to determine the frequencies of these variables. The project started with the initial importation of the data into R Studio and loading in the tidyverse and janitor packages to ensure the data was clean and ready to be analyzed. This consisted of creating a four-by-two frequency table for each color of ball with there being 307 blue balls, 192 green balls, 408 red balls, and 93 yellow balls. A five-by-two frequency table was also created to represent the different counts for each ball label with label A having 130 balls, label B having 305 balls, label C having 59 balls, label D having 325 balls, and label E having 181 balls. A bar chart was created to show the variation in the different color counts of balls, which visibly shows red a lot higher than yellow, and yellow being a lot lower than blue or green.



Furthermore, a bar chart representing the different labels was created which visibly presents the labels B and D as a lot higher than the other three labels. Label C is also shown as having a much

smaller count than any of the other four labels.



After importing the data, cleaning it, and making it presentable, the project shifted towards probability of drawing different colored balls or different labeled balls. The first scenario presented was the probability of drawing a green ball which was stored as prob6_result and was 0.192. After that the probability of drawing both a blue or a red ball was 0.715 and means there is a fairly high chance (almost 3 out of every 4 times) that if a ball was to be randomly selected it would be blue or red. This was stored as prob7_result. Shifting the focus to label probability, the chance that a ball drawn has the label A or C is 0.189 and shows the combination of the two least-counted labels out of all five labels involved in this dataset – this was stored as prob8_result. The probability of drawing a yellow ball or a ball with a D combined the variables label and color to give us a result of 0.036 which is a fairly low chance of getting both of these characteristics when randomly choosing a ball. This result was stored as prob9_result. In a

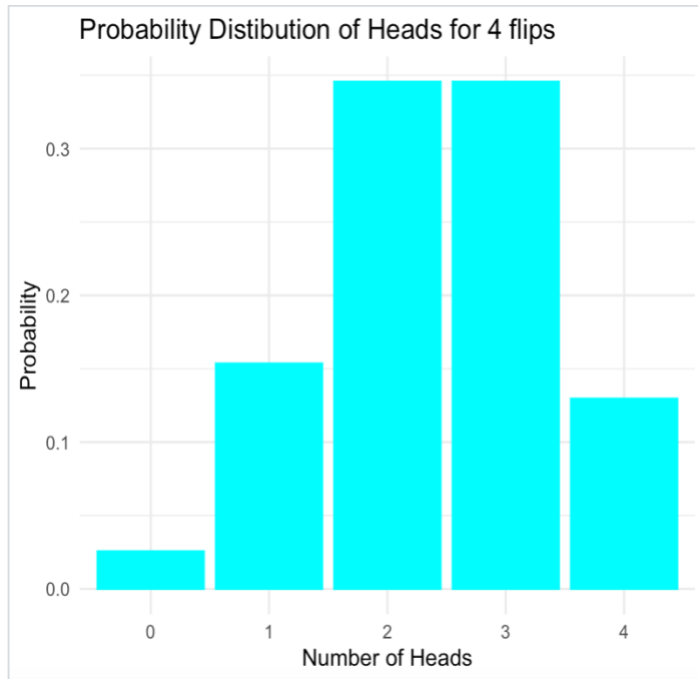
similar fashion, the probability of drawing a yellow ball or a ball with a D was determined to be 0.382 and was stored as `prob10_result`.

As the dataset focus changed to probability without replacement, the chance of drawing a blue ball followed by a red ball without replacement ended up being 0.1253814 or roughly one in every eight times this would occur. This result was stored as `prob11_result` in the R Studio script. Next, the probability of drawing four green balls in a row without replacement was determined to be 0.001324826 which is a very low number, meaning the likelihood of this is very small (around a 0.1% chance) and it was stored as `prob12_result`. In the last analysis of probability without replacement, the probability of drawing a red ball followed by a ball with a B without replacement was determined to be 0.124564 which is also around one in every eight times and this was stored as `prob13_result`. A function was also written called `my_factorial` which computes the factorial of a given number. A factorial is the product of multiplying a whole number by every positive integer below it down to the value 1 – it can be thought of as total number ways that items can be arranged. So, for the number 3, the factorial would be 6 because there are six ways to arrange three items and for the number 5, the factorial is 120 because there are one hundred twenty ways to arrange five items. This dataset was ultimately focused on the different colors and labels associated with picking balls, as well as the probability that these specific colors and labels would be chosen with and without replacement.

For the second part of the dataset, the focus was shifted to creating a coin flipping data frame while considering an unfair coin that has a probability of 0.6 of landing on heads. A sixteen by four data frame called `coin_outcomes` was created to show all the possible outcomes of flipping the coin in this situation four times. This variable `coin_outcomes` added a section to compute the probability of each row outcome. For example, the probability of drawing four

heads would be 0.1296 and the probability of drawing three heads with your first three draws and then a tails is lower at a value of 0.0864. The lowest probability in this entire dataset would be drawing four tails with a value of 0.0256, so only occurring approximately 2.5 percent of the time. There would be five possible outcomes in the dataset if the number of heads were counted in each row. Therefore, a new variable called `num_heads_prob` was created to show the probability of each of the five possible outcomes. For example, the probability of zero heads is 0.0256, one head is 0.1536, two heads are 0.3456, three heads are 0.3456, four heads are 0.3456, and five heads are 0.1296. The probability of getting either two or three heads in flipping this coin would occur almost seventy percent of the time and getting zero heads is a very unlikely event since it only occurs around 2.5% of the time.

Similar to the ball analysis, the probability of different coin flip outcomes was analyzed. For example, the probability of an outcome of three heads was stored as `prob18_result` and was determined to be 0.3456. The chance of getting an outcome of two heads or four heads was calculated to be 0.4752 which occurs almost half of the time and was stored as `prob19_result` in the R Studio script. The probability of an outcome of less than or equal to three heads was stored as `prob20_result` and is 0.8704 – it would be pretty likely if when flipping the coin in this analysis, an individual would get an outcome of less than or equal to three heads around eighty seven percent of the time. After calculating all of the different probabilities for coin flips, a bar chart was created for the probability distribution of heads for four flips. The chart shows the audience that visually they would be likely to get 2 or 3 heads each time they did four coin flips at a percentage that is greater than thirty. On the other hand, they would be a lot less likely to get heads 0 times if they did four coin flips because probability of that would be under five percentage just from what the bar chart shows.



Getting heads 2 or 4 times made up the middle part of the probability distribution of heads for 4 flips and these values were both in between 0.1 and 0.2. The dataset was ultimately focused on flipping a coin a different number of times and seeing the probability of a certain result, such as heads for a specific number of times.

The last part of the data considered a soccer team with a 75% chance of winning a game at home and a 50% chance of winning away games. The team was about to play ten upcoming games with five being at home and five being away. The variable `prob22_result` looked at the chance that the team would win exactly ten games, and this is 0.07415771 meaning there is around a 7.5% chance that this team would win all five of their upcoming home and away games. The probability that the team will win more than one game is incredibly high and is 0.9998474. It would be incredibly unlikely, less than a 0.001 (0.1%) chance that the team would not win more than one game in this upcoming ten game series. This result was stored as a variable `prob23_result`. For the variable `prob24_result`, the question “how many different ways

could you pick five games at random and have three home games and two away games” was considered and this ended up being 100. This last part of the data was focused on analyzing the probability of this soccer team winning their next ten games at home and away.

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